Exercise Sheet 9

Diffusion and Epidemics

5942UE Network Science

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9.A Diffusion as a Network Process

Exercise 9.A.1: Simple Diffusion Trace

Directed graph: $A \to B, A \to C, B \to D, C \to D, D \to E$.

Node A is active at $t = 0$. At each step, an active node activates all out-neighbours.

1. Trace the active set at $t = 1, 2, 3$.
2. Which node is never reached? Why?
3. If you add the edge $B \to C$, does anything change for this seed?

Solution:

1. $t = 0$: A. $t = 1$: A, B, C. $t = 2$: A, B, C, D. $t = 3$: A, B, C, D, E.

2. All nodes are reachable from A. (An isolated node with no incoming edges from this set would never be reached.)

3. Adding $B \to C$ changes nothing — C is already activated at $t = 1$ via $A \to C$. Redundant paths don't accelerate spread beyond what existing paths provide.

Exercise 9.A.2: Structural Bottlenecks

Two 8-node graphs with seed S:

- **Star:** S connected to every other node.
- **Line:** $S - A - B - C - D - E - F - G$.

1. How many steps until all nodes are active in each?
2. Where is the structural bottleneck?
3. A single edge fails. Which topology is more affected?

Solution:

1. Star: 1 step. Line: 7 steps.

2. The line has a bottleneck at every edge — removing any node severs everything downstream. The star has no internal bottleneck.

3. Removing one edge from the line blocks up to 6 of 7 nodes. Removing one edge from the star isolates exactly one node. The line is vastly more vulnerable.

Exercise 9.A.3: Centrality and Diffusion Speed

Load the karate club graph and BFS-diffuse from the highest-degree node.

1. How many steps until all 34 nodes are active?
2. Repeat from the lowest-degree node. How does the curve change?
3. What does this say about hubs?

Solution:

```python
import networkx as nx

G = nx.karate_club_graph()
cent = nx.degree_centrality(G)
seed = max(cent, key=cent.get)

active, frontier = {seed}, {seed}
for t in range(8):
    new_frontier = {nbr for node in frontier for nbr in G.neighbors(node)
                    if nbr not in active}
    active |= new_frontier
    frontier = new_frontier
    print(f"t={t+1}: {len(active)} active")
```

A hub seed reaches the full graph in 3–4 steps. The lowest-degree node takes 5–6+ steps with a much flatter early curve. Superspreaders have outsized impact because they bridge to the rest of the network in fewer hops.

Exercise 9.A.4: Paths and Reachability

Company-wide email cascade from CEO (C) must reach 500 employees. The network has a core-periphery structure: 50 managers in a dense core, each linked to ≈ 10 peripheral employees.

1. Does a message from C to a peripheral employee require passing through the core?
2. If 5 of 50 managers are absent, what fraction of periphery is potentially unreachable?
3. How could the company become more resilient?

Solution:

1. Yes — peripheral nodes only link to core nodes, so every path passes through the core.

2. Each absent manager isolates ≈ 10 peripheral nodes → about $50/450 \approx 11$ unreachable if no redundant paths exist.

3. Add cross-edges between peripheral employees; give each peripheral employee two or more core contacts; identify high-betweenness bridge managers and ensure backup coverage. Resilience requires *redundant paths*.

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9.B Epidemic Models and Simple Contagion

Exercise 9.B.1: SIR Trace

SIR rules (synchronous): $S \to I$ if any infected neighbour, with probability β. $I \to R$ at the next step (single-step infectious period).

Path graph $A - B - C - D - E$ with B infected at $t = 0, \beta = 1.0$.

1. Trace each node's state at $t = 0, 1, 2, 3, 4$.
2. When does E become infected?
3. With $\beta = 0.5$, what is $P(\text{C not infected at } t = 1)$?

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Solution:

1. State trace:

Time	A	B	C	D	E
$t = 0$	S	**I**	S	S	S
$t = 1$	**I**	**R**	**I**	S	S
$t = 2$	**R**	R	**R**	**I**	S
$t = 3$	R	R	R	**R**	**I**
$t = 4$	R	R	R	R	**R**

2. E is infected at $t = 3$, recovers at $t = 4$.

3. C has one infected neighbour (B), so it is infected with probability 0.5. $P(\text{not infected}) = 0.5$.

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Exercise 9.B.2: Structural Effect of Topology

SIR with $\beta = 1.0$, one-step recovery, seed = node 1.

- **Line L:** $1 - 2 - 3 - 4 - 5 - 6$
- **Star S:** node 1 is the hub of $2, 3, 4, 5, 6$

1. Maximum simultaneously infected in L?
2. Maximum simultaneously infected in S?
3. Which produces a sharper epidemic peak? Implications?

Solution:

1. Line: at any time only 1 node is infected (a moving wave).

2. Star: at $t = 1$, all 5 spokes are infected simultaneously.

3. The star produces a sharper, taller peak. **High-degree hubs create epidemic explosions** — a single infected hub exposes all neighbours at once. This is why superspreader events are dangerous and why targeted vaccination of hubs is so effective.

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Exercise 9.B.3: SIR Simulation in NetworkX

Implement synchronous SIR on the karate club graph and analyse it.

1. At what step does the epidemic peak?
2. What fraction ends in R (ever infected)?
3. Vary β from 0.1 to 0.8. Where does the epidemic reliably infect > 80?

Solution:

```python
import networkx as nx
import random

def sir_simulation(G, seed_node, beta=0.3, steps=20):
    state = {n: 'S' for n in G.nodes()}
    state[seed_node] = 'I'
    history = []
    for _ in range(steps):
        new_state = state.copy()
        for node in G.nodes():
            if state[node] == 'I':
                new_state[node] = 'R'
            elif state[node] == 'S':
                inf_nbrs = [n for n in G.neighbors(node) if state[n] == 'I']
                if any(random.random() < beta for _ in inf_nbrs):
                    new_state[node] = 'I'
        state = new_state
        history.append(tuple(sum(1 for s in state.values() if s == k)
                             for k in 'SIR'))
    return history
```

With $\beta = 0.3$ and a hub seed, the peak hits around step 3–5. Below $\beta \approx 0.1$ the epidemic dies out; above $\beta \approx 0.4$ nearly the whole network is eventually infected. The dense community structure makes the karate club highly susceptible once the threshold $R_0 > 1$ is crossed.

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Exercise 9.B.4: Simple vs. Complex Contagion

For each scenario, decide whether it is **simple** (one contact sufficient) or **complex** (multiple confirmations needed):

(a) A cold virus. (b) Adopting a new software tool only after three colleagues recommend it. (c) Sharing a news article posted by one friend. (d) Joining a protest after pressure from multiple community members. (e) Updating a belief after seeing a claim from several independent sources.

For (b) and (d), how must network structure differ from simple contagion?

Solution:

Scenario	Type
(a) Cold	Simple
(b) Software	Complex
(c) Article	Simple
(d) Protest	Complex
(e) Belief	Complex

For (b) and (d) to spread, the network needs **dense local clusters** where multiple neighbours have already adopted. A bridge or weak tie — which excels for simple contagion — fails for complex contagion because the node on the other side has only one adopter neighbour.

Complex contagion spreads through tight communities and stalls at bridges — exactly the opposite of simple contagion, which exploits bridges for rapid long-range diffusion.

Wrap-Up

- diffusion is bounded by reachable paths; redundant paths don't help unless primary paths fail
- bottlenecks and hubs control epidemic dynamics
- the same β produces dramatically different curves on different topologies
- complex contagion reverses the role of bridges and clusters compared to simple contagion